Query Campus: A Retrieval-Augmented Generation Based Conversational Assistant For College Website Information Access

A Report submitted
in partial fulfillment for the degree of

Bachelor of Engineering

in

Information Technology

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CERTIFICATE

The project report entitled “Query Campus: A Retrieval-Augmented Generation Based Conversational Assistant For College Website Information Access” submitted by Akshat Sharma, Tanmay Choudhary, Riya Rawat is approved for partial fulfillment for the award of Bachelor of Engineering in Information Technology degree by Devi Ahilya Vishwavidyalaya, Indore.

Internal Examiner External Examiner

Recommendation

The dissertation entitled “Query Campus: A Retrieval-Augmented Generation Based Conversational Assistant For College Website Information Access” submitted by Akshat Sharma, Tanmay Choudhary, Riya Rawat is a satisfactory account of the bonafide work done under my supervision and is recommended towards the partial fulfillment for the award of Bachelor of Engineering in Information Technology degree by Devi Ahilya Vishwavidyalaya, Indore.

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Candidate Declaration

We hereby declare that the work which is being presented in this project entitled "Query Campus: A Retrieval-Augmented Generation Based Conversational Assistant For College Website Information Access" in partial fulfillment of degree of Bachelor of Engineering in Information Technology is an authentic record of our own work carried out under the supervision and guidance of Vedpriya Dongre, [Designation of Project Guide] in Department of Computer Engineering, Institute of Engineering and Technology, Devi Ahilya Vishwavidyalaya, Indore.

We are fully responsible for the matter embodied in this project in case of any discrepancy found in the project and the project has not been submitted for the award of any other degree.

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SELF-DECLARATION OF PLAGIARISM REPORT

The project report entitled “Query Campus: A Retrieval-Augmented Generation Based Conversational Assistant For College Website Information Access” submitted by Akshat Sharma, Tanmay Choudhary, Riya Rawat in partial fulfillment of the requirement for the award of the degree of B.E. in Information Technology has been checked for plagiarism using a standard plagiarism detection tool.

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ABSTRACT

Information about a college is spread over many web pages, PDF notices and a separate timetable website. Students lose time searching for admission rules, fees, teacher details and class timetables, and general chatbots often give wrong answers about a college. Query Campus is a chat assistant that answers questions about the Institute of Engineering & Technology, DAVV, Indore in simple English. It uses only the information published by the institute and shows the pages it used.

The system collects 1,009 documents, cuts them into 2,864 small chunks and stores them in a Pinecone database. For each question, the backend searches by meaning and by keywords, picks the best 16 chunks and asks Google Gemini to answer using only those chunks. The answer appears word by word in a React chat screen, with source links and downloadable documents. Backup models, time limits and a limit on questions per visitor keep the website reliable and safe.

On a test set of 18 real questions, the assistant answered all correctly, usually in 1.5 to 4 seconds, and it said it could not find the answer when the answer was not there. The project shows that a small, low-cost system can give a college a reliable question-answering service.

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Chapter-1 Introduction

1.1 Overview and Issues Involved

A college website is the first place where students look for information. They want to know about admission, fees, scholarships, exams and class timetables. But this information is spread over many pages and files. The IET DAVV website has 160 web pages and 606 PDF files. The class timetables are on a different website, which gives us 243 more documents. To find one simple fact, a student has to guess which page or file to open.

AI chatbots such as Google Gemini can answer questions in simple language. But they have not read the college notices, and they often give wrong answers with full confidence. A wrong fee or date can cost a student time and money. This project uses a method called Retrieval-Augmented Generation (RAG). The system first finds the right text on the college website. Then the AI writes an answer using only that text.

The main issues are:

- **Scattered information.** Facts are inside web pages, PDF files and a separate timetable website.
- **Names and numbers.** Search by meaning is weak for exact names and codes, so we also need keyword search.
- **Trust.** Answers must show their sources, and the system must say “I could not find it” when the answer is missing.
- **Old data.** Notices change every semester, so the data must be easy to update.
- **Free plans.** Our online services allow only a few requests per minute, so the system must handle delays and failures.

1.2 Problem Definition

Students cannot ask a question in simple language and get a correct answer from the college website. The information is hard to find, and general chatbots do not know the college data and may give wrong answers.

We need a website where a student can type a question and get a short, correct answer with its source. The objectives are:

1. Collect the text of the college website and the timetable website.
2. Store the text so that it can be searched by meaning and by keywords.
3. Write answers with an AI that uses only the college text, and show the sources.
4. Show the answer in a simple chat screen that also gives documents such as timetables.
5. Test the accuracy with a fixed list of questions.

1.3 Proposed Solution

Query Campus is a chat assistant for IET DAVV. It works in two parts, shown in Fig. 1.1. In the **indexing part**, the operator collects the pages, cuts them into small chunks and saves them in a database. In the **answering part**, the system searches the database for the best chunks, sends them to Gemini, and shows the answer to the student with its sources.

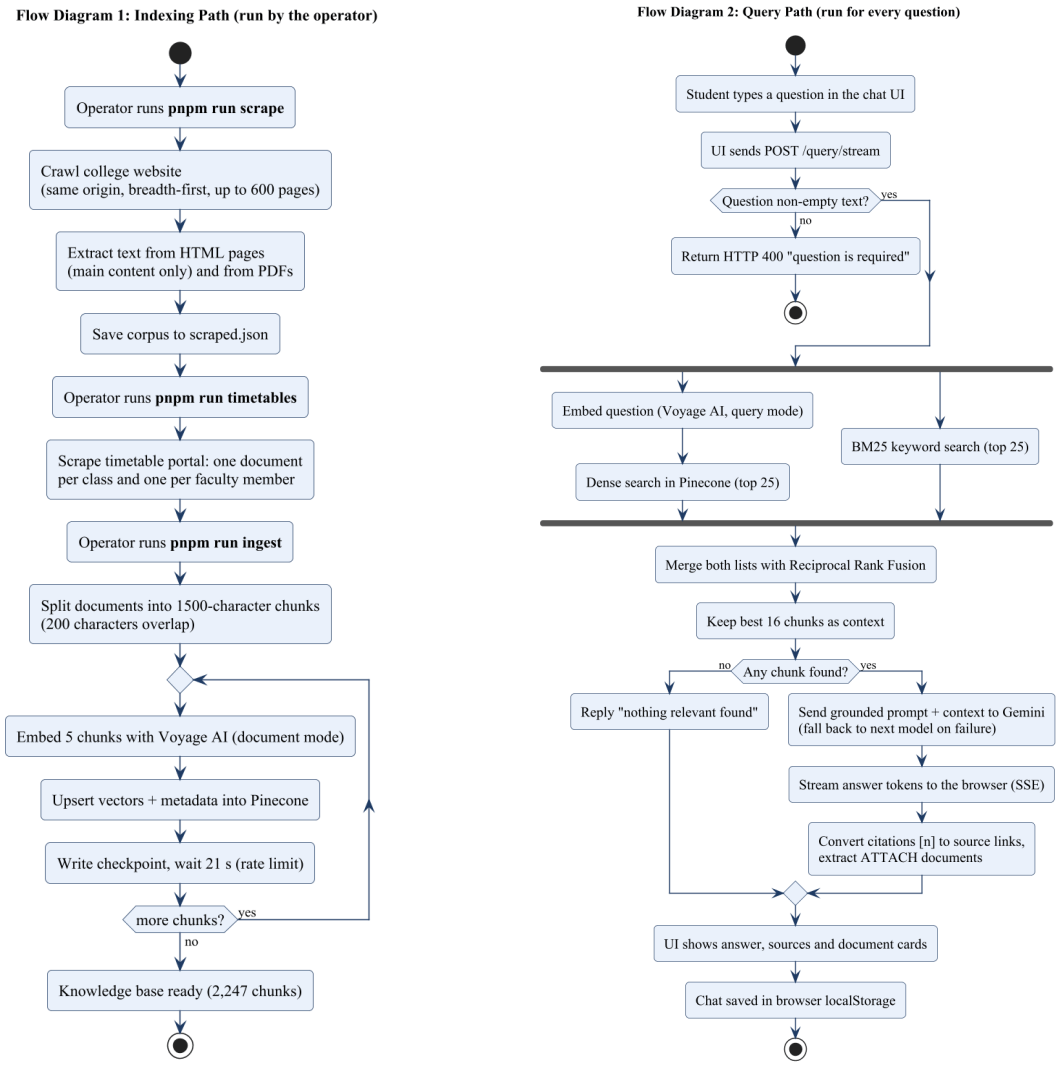


Fig. 1.1: Flow diagrams of Query Campus (left: indexing, right: answering)

Functions of the project:

- Collects text from the college website (web pages and PDFs) and from the timetable website.
- Saves the text as small chunks in a Pinecone database.
- Finds the best chunks by meaning and by keywords.
- Writes the answer with Gemini, with sources, and says so when it cannot find the answer.
- Shows the answer in a chat screen with saved chats, source links and document cards.
- Works safely for the public: a limit on questions per visitor and a limit on question length.

- Has a test tool with 18 questions to check the accuracy.

Why it is better than existing solutions. The college website search cannot understand questions. FAQ chatbots only answer fixed questions. General chatbots make up facts. Paid college chatbots are closed and are not made for IET DAVV. Query Campus answers only from the college pages, shows its sources, reads the timetable website too, says “I could not find it” instead of guessing, and is easy to update.

Chapter-2 Literature Survey

2.1 Existing Solutions

This section describes the existing ways to find information on a college website. Table 2.1 compares them with Query Campus.

(a) The IET DAVV website. It has admissions, faculty, syllabus, notices and downloads. It is the most trusted source, but related facts are in different places and many are only inside PDF files. A user cannot ask a question directly.

(b) The timetable website. The user selects the session, semester and course, and the site shows the timetables. It is accurate, but it cannot answer "Which classes does this teacher take?" without opening many timetables.

(c) Perplexity AI. Perplexity [1] is a chat-style search engine that shows links with its answers. But it searches the whole web, so a college answer may come from an unofficial or old page.

(d) ChatPDF. ChatPDF [2] answers questions about one uploaded PDF. The college has more than 600 PDFs, so a student would have to find and upload the right one.

(e) Paid college chatbots. Ivy.ai [3], Ocelot [4], Element451 BoltBot [5] and Gravyty [6] sell chatbots to universities. They work well, but they are closed, paid products that are set up by the company.

(f) General chatbots and FAQ bots. ChatGPT and Gemini have not read the current college notices and may give wrong answers. FAQ bots only reply to a fixed list of questions.

Table 2.1: Comparison of existing solutions with Query Campus

Solution	What it does	Problem	How Query Campus is better
IET DAVV website	Pages, notices and downloads	Information is scattered; no direct answers	Gives a direct answer with its source

Solution	What it does	Problem	How Query Campus is better
Timetable website	Shows class timetables	User must know the class; cannot ask who teaches what	Answers questions about teachers, subjects and coordinators
Perplexity AI [1]	Web search with a chat answer	Uses the whole web, not only official pages	Uses only official IET DAVV pages
ChatPDF [2]	Questions on one uploaded PDF	User must upload each file	All college pages are ready in advance
Ivy.ai, Ocelot, BoltBot, Gravyty [3]-[6]	Paid chatbots for colleges	Closed, paid, not made for IET DAVV	Open and works on free plans
General chatbots (ChatGPT, Gemini)	Open conversation	No college data; may make up facts	Answers only from college text
FAQ chatbots	Fixed questions and replies	Fail on new questions	Accepts any question

Gap. No existing solution gives a direct answer that comes only from the official IET DAVV website and timetable website, shows its source, and can be updated cheaply. Query Campus fills this gap.

2.2 Methodology

2.2.1 Steps followed

1. **Plan.** We wrote the synopsis and the SRS to decide what the project must do.
2. **Collect data.** We built a crawler for the college website and a scraper for the timetable website.
3. **Index.** We cut the text into chunks, converted them into numbers and saved them in a database.
4. **Answer.** We built the search, the Gemini prompt and the streaming answer.

5. **Chat screen.** We built the React chat screen with sources, saved chats and document cards.
6. **Test and improve.** We tested with 18 real questions and fixed the problems we found.

2.2.2 Concepts needed to understand the project

Crawler. A program that opens a web page, collects its links and repeats this for every new link. Ours stays on the college website.

Chunk. A small piece of a document (about 1,500 characters). Small pieces are easier to search than a whole document.

Embedding. A list of numbers that represents the meaning of a text. Texts with similar meaning get similar numbers.

Vector database. A database that stores embeddings and quickly finds the ones closest to a question. We use Pinecone.

Keyword search. Finds texts that contain the exact words of the question. It is good for names and codes. We use it together with search by meaning and merge the two result lists.

Large language model (LLM). An AI that writes text. We use Google Gemini.

Grounding. Telling the AI to use only the given text, to show its sources, and to say when the answer is missing. This stops made-up answers.

RAG. Retrieval-Augmented Generation: first find the right text, then let the AI write the answer from it. To update the knowledge, we only update the database.

Streaming. The server sends the answer in small parts, so the student sees the words appear while the answer is being written.

Chapter-3 Analysis & Design

3.1 Software Requirements

Table 3.1 lists the software we use and why. We also need three online services (Voyage AI, Pinecone and Google Gemini) with their API keys, and public access to the college website.

Table 3.1: Software requirements

Part	Technology	Why we use it
Backend	Node.js, NestJS, TypeScript	One language everywhere; NestJS keeps the code in clear modules.
Web crawling	Axios, Cheerio, pdf-parse	Download pages, read their text, and read text from PDF files.
Keyword search	MiniSearch	Fast search for exact words such as names and codes.
Embeddings	Voyage AI (voyage-3)	Turns text into numbers; no GPU needed.
Database	Pinecone	Stores the numbers and finds the closest ones quickly.
AI answers	Google Gemini	Fast and low cost; backup models are available.
Frontend	React 19, Vite, Tailwind CSS	Fast, responsive chat screen.
Testing	Vitest, Supertest, oxlint	Unit tests, end-to-end tests and code checks.
Project tools	pnpm workspaces, .env files	One repository for both apps; secret keys stay out of the code.

Main requirements. The system must collect and save the college text, answer questions with sources, give documents, keep chat history in the browser and report its health. It must answer within a few seconds, keep working when an online service is slow, keep secret keys on the server, and work on phones.

3.2 Hardware Requirements

Query Campus needs no special hardware. The heavy work is done by online services, so ordinary computers are enough (Table 3.2).

Table 3.2: Hardware requirements

Item	Minimum	Reason
Processor	Dual-core	The heavy work is done by online services.
Memory	4 GB (8 GB for development)	The search index and the tools need a few hundred MB.
Storage	1 GB free	Collected text, a search copy (about 4 MB) and project files.
Internet	Broadband	Every question uses three online services.
User device	Any phone or computer with a modern browser	The screen works on small and large displays.
Server	1 CPU, 1 GB RAM	Enough for one college; the server keeps no user data.

3.3 Analysis Diagrams

3.3.1 Use Case Model

Fig. 3.1 shows who uses the system. The **user** (student, applicant or staff) asks questions, reads answers, gets documents and manages chats. The **operator** collects and updates the data and runs the tests. Outside systems also take part: the college website and timetable website (data), Voyage AI (numbers for text), Pinecone (database) and Gemini (answers).

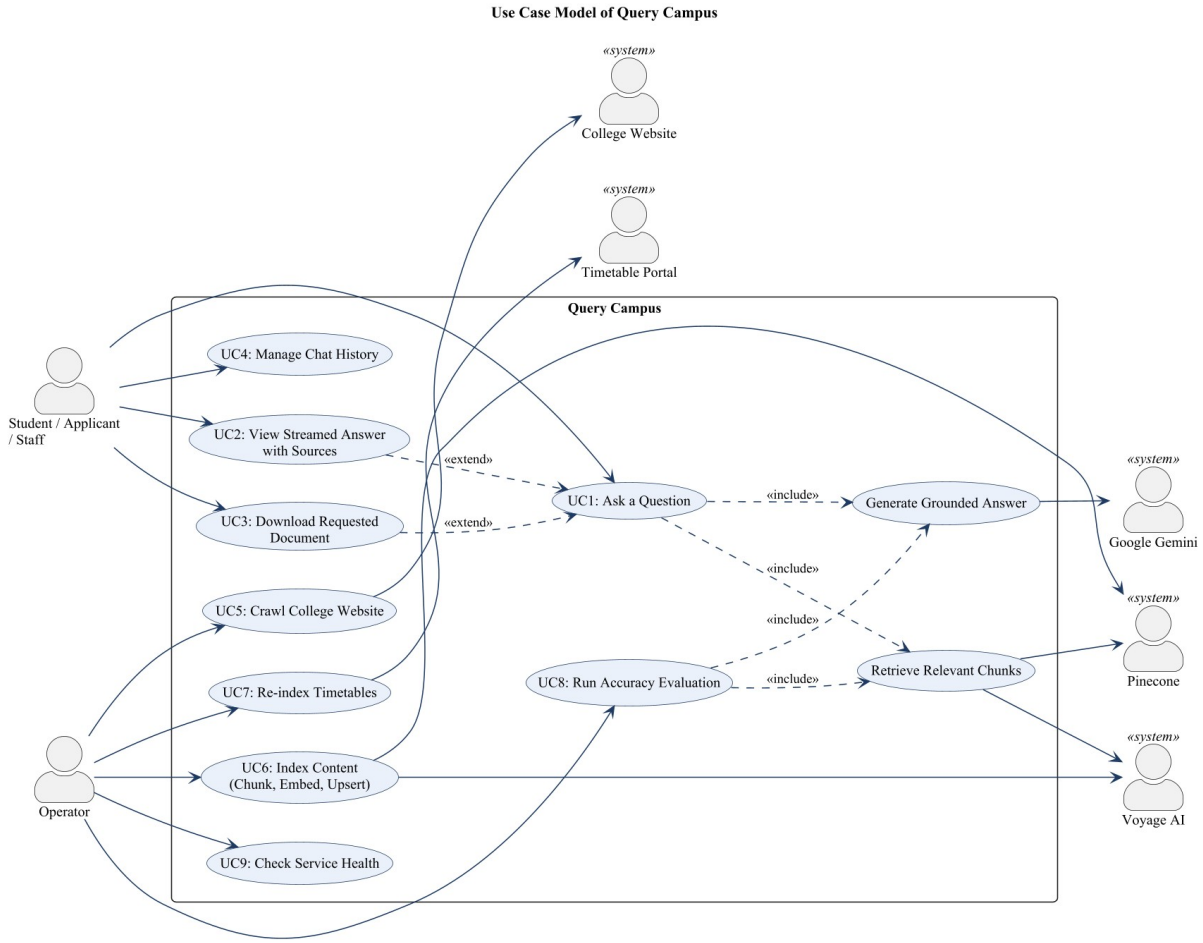


Fig. 3.1: Use case model of Query Campus

3.3.2 Use Case Description

Each use case is described below in the Rational format. The numbers have the form 3.3.2.n.k, where n is the use case and k is the item.

3.3.2.1 UC1: Ask a Question

No.	Item	Description
3.3.2.1.1	Use Case Name	UC1: Ask a Question
3.3.2.1.2	Brief Description	The user types a question and gets a short answer based on the college website.
3.3.2.1.3.1	Basic Flow	1. The user types a question and presses Enter. 2. The screen sends the question to the backend. 3. The system finds the best chunks (UC10).

No.	Item	Description
		4. The system writes the answer (UC11). 5. The answer appears word by word and is saved in the chat.
3.3.2.1.3.2	Alternative Flows	A1 The question is blank or longer than 500 characters: the system rejects it with a message. A2 Nothing relevant is found: the system says it could not find an answer. A3 A service fails: the screen shows a short error message. A4 The user sends more than 20 questions in a minute: the system asks the user to wait.
3.3.2.1.4	Special Requirements	The answer must start within a few seconds. No login is needed.
3.3.2.1.5	Preconditions	The backend is running and the database is filled.
3.3.2.1.6	Post Conditions	The answer is shown and saved in the user's browser. Nothing is stored on the server.
3.3.2.1.7	Extension Points	View Answer with Sources (UC2); Download Requested Document (UC3).

3.3.2.2 UC2: View Streamed Answer with Sources

No.	Item	Description
3.3.2.2.1	Use Case Name	UC2: View Streamed Answer with Sources
3.3.2.2.2	Brief Description	The user reads the answer and opens the pages it came from.
3.3.2.2.3.1	Basic Flow	1. The answer appears as it is written. 2. When it is complete, the sources appear under it as links. 3. The user clicks a link to open the original page.
3.3.2.2.3.2	Alternative Flows	A1 The answer has no sources: no source links are shown.
3.3.2.2.4	Special Requirements	The same source is never shown twice.

No.	Item	Description
3.3.2.2.5	Preconditions	A question has been sent (UC1).
3.3.2.2.6	Post Conditions	The full answer and its links are saved with the chat.
3.3.2.2.7	Extension Points	None.

3.3.2.3 UC3: Download Requested Document

No.	Item	Description
3.3.2.3.1	Use Case Name	UC3: Download Requested Document
3.3.2.3.2	Brief Description	When the user asks for a document such as a timetable, a document card with Open and Download buttons is shown.
3.3.2.3.3.1	Basic Flow	1. The user asks, for example, “Send me the 4th year CSE section B timetable”. 2. The system finds the document and shows a document card. 3. The user clicks Open to view it, or Download to save it.
3.3.2.3.3.2	Alternative Flows	A1 The link is not a PDF of the college website: the download is refused. A2 The college website cannot be reached: the system shows a short message.
3.3.2.3.4	Special Requirements	Only PDF files from the college website can be downloaded.
3.3.2.3.5	Preconditions	The document is in the database.
3.3.2.3.6	Post Conditions	The user has opened or saved the document.
3.3.2.3.7	Extension Points	None.

3.3.2.4 UC4: Manage Chat History

No.	Item	Description
3.3.2.4.1	Use Case Name	UC4: Manage Chat History
3.3.2.4.2	Brief Description	The user starts new chats, opens old chats and deletes chats.
3.3.2.4.3.1	Basic Flow	1. The user clicks “New chat” or picks an old chat in the side panel. 2. The screen shows that chat. 3. The user can delete a chat with the cross button.
3.3.2.4.3.2	Alternative Flows	A1 The browser blocks local storage: chats work but are lost when the page is reloaded.
3.3.2.4.4	Special Requirements	Chats are saved only in the user’s browser.
3.3.2.4.5	Preconditions	The website is open in a browser.
3.3.2.4.6	Post Conditions	The chat list is updated.
3.3.2.4.7	Extension Points	None.

3.3.2.5 UC5: Crawl College Website

No.	Item	Description
3.3.2.5.1	Use Case Name	UC5: Crawl College Website
3.3.2.5.2	Brief Description	The operator collects the text of all pages and PDF files of the college website.
3.3.2.5.3.1	Basic Flow	1. The operator runs “pnpm run scrape”. 2. The crawler opens pages one after another and follows links on the same website. 3. It saves the text of each page and PDF. 4. It writes everything into one file.
3.3.2.5.3.2	Alternative	A1 A page cannot be opened: it is skipped and the crawl

No.	Item	Description
	Flows	continues.
3.3.2.5.4	Special Requirements	The crawler is polite: four pages at a time, with time limits.
3.3.2.5.5	Preconditions	The operator has set up the project.
3.3.2.5.6	Post Conditions	The file scraped.json holds the collected text.
3.3.2.5.7	Extension Points	None.

3.3.2.6 UC6: Index Content

No.	Item	Description
3.3.2.6.1	Use Case Name	UC6: Index Content
3.3.2.6.2	Brief Description	The operator saves the collected text in the database so it can be searched.
3.3.2.6.3.1	Basic Flow	1. The operator runs “pnpm run ingest” (or adds --incremental for new pages only). 2. The system cuts the text into chunks. 3. It converts the chunks into numbers with Voyage AI. 4. It saves them in Pinecone, a few at a time, and waits when the free limit requires it.
3.3.2.6.3.2	Alternative Flows	A1 A service is busy: the system waits and tries again. A2 The run stops: the next run continues from where it stopped.
3.3.2.6.4	Special Requirements	Running it again does not create duplicates.
3.3.2.6.5	Preconditions	scraped.json exists (UC5) and the API keys are set.
3.3.2.6.6	Post Conditions	All chunks can be searched.
3.3.2.6.7	Extension Points	None.

3.3.2.7 UC7: Re-index Timetables

No.	Item	Description
3.3.2.7.1	Use Case Name	UC7: Re-index Timetables
3.3.2.7.2	Brief Description	The operator updates the class and teacher timetables each semester.
3.3.2.7.3.1	Basic Flow	1. The operator runs “pnpm run timetables”. 2. The system reads all timetables of the latest session. 3. It makes one document per class and one per teacher. 4. It replaces the old timetable chunks in the database.
3.3.2.7.3.2	Alternative Flows	A1 With “--dry-run” the documents are only saved to a file.
3.3.2.7.4	Special Requirements	Only timetable chunks are replaced.
3.3.2.7.5	Preconditions	The timetable website is reachable.
3.3.2.7.6	Post Conditions	The database has the current timetables.
3.3.2.7.7	Extension Points	None.

3.3.2.8 UC8: Run Accuracy Test

No.	Item	Description
3.3.2.8.1	Use Case Name	UC8: Run Accuracy Test
3.3.2.8.2	Brief Description	The operator checks the quality of search and answers with a fixed list of questions.
3.3.2.8.3.1	Basic Flow	1. The operator runs “pnpm run eval”. 2. For each question the system checks that the right page was found and the right facts are in the answer. 3. It prints pass or fail for each question and compares with the last run.
3.3.2.8.3.2	Alternative	A1 With “--retrieval-only” only the search is tested.

No.	Item	Description
	Flows	
3.3.2.8.4	Special Requirements	Question numbers are saved, so repeated runs are quick.
3.3.2.8.5	Preconditions	The database is filled.
3.3.2.8.6	Post Conditions	A report of the run is saved.
3.3.2.8.7	Extension Points	None.

3.3.2.9 UC9: Check Service Health

No.	Item	Description
3.3.2.9.1	Use Case Name	UC9: Check Service Health
3.3.2.9.2	Brief Description	The operator checks that the backend is running.
3.3.2.9.3.1	Basic Flow	1. The operator opens /health. 2. The system answers “ok”.
3.3.2.9.3.2	Alternative Flows	A1 The server is down: there is no answer.
3.3.2.9.4	Special Requirements	No login is needed.
3.3.2.9.5	Preconditions	None.
3.3.2.9.6	Post Conditions	None.
3.3.2.9.7	Extension Points	None.

3.3.2.10 UC10: Retrieve Relevant Chunks (included)

No.	Item	Description
3.3.2.10.1	Use Case	UC10: Retrieve Relevant Chunks (included)

No.	Item	Description
	Name	
3.3.2.10.2	Brief Description	The system finds the chunks that best match the question.
3.3.2.10.3.1	Basic Flow	1. The system searches by meaning and by keywords at the same time. 2. It merges the two result lists. 3. It keeps the best 16 chunks, with at most three from one page.
3.3.2.10.3.2	Alternative Flows	A1 One of the two searches is not working: the other one is used.
3.3.2.10.4	Special Requirements	For questions about a person, faculty pages come first.
3.3.2.10.5	Preconditions	The database is filled.
3.3.2.10.6	Post Conditions	A list of chunks is ready.
3.3.2.10.7	Extension Points	None.

3.3.2.11 UC11: Generate Grounded Answer (included)

No.	Item	Description
3.3.2.11.1	Use Case Name	UC11: Generate Grounded Answer (included)
3.3.2.11.2	Brief Description	The system asks Gemini to write the answer using only the chunks.
3.3.2.11.3.1	Basic Flow	1. The system sends the chunks and the question to Gemini. 2. Gemini writes the answer, and the words are sent to the screen as they come. 3. The system turns the citation numbers into source links.
3.3.2.11.3.2	Alternative Flows	A1 Gemini is busy: the system tries another Gemini model.
3.3.2.11.4	Special	Gemini must not use facts outside the chunks.

No.	Item	Description
	Requirements	
3.3.2.11.5	Preconditions	Chunks were found.
3.3.2.11.6	Post Conditions	The answer and its sources are ready.
3.3.2.11.7	Extension Points	None.

3.4 Design Diagrams

3.4.1 Architecture Diagram

System overview

Students, applicants and staff use the chat screen. The operator runs the commands that collect and update the data. The college web team publishes the pages we read; we never change them. Voyage AI, Pinecone and Google give us numbers for text, storage and answers. In short: the question goes from the browser to the backend, the backend finds the best text and asks Gemini to write the answer, and the answer comes back with links to the college pages.

Architecture strategy

- **Layers.** A screen layer, an application layer, an integration layer and a separate indexing layer.
- **Offline and online parts.** Slow work (collecting and indexing) is done in advance, so answering is fast.
- **Two kinds of search.** Search by meaning finds the topic and keyword search finds exact names. If one fails, the other still works.
- **Answers from evidence only.** Gemini sees only the found text, and the server creates the source links.
- **Safe for the public.** Secret keys stay on the server, each visitor can ask 20 questions a minute, only allowed websites can call the API, and nothing about visitors is stored.

Block diagram and explanation

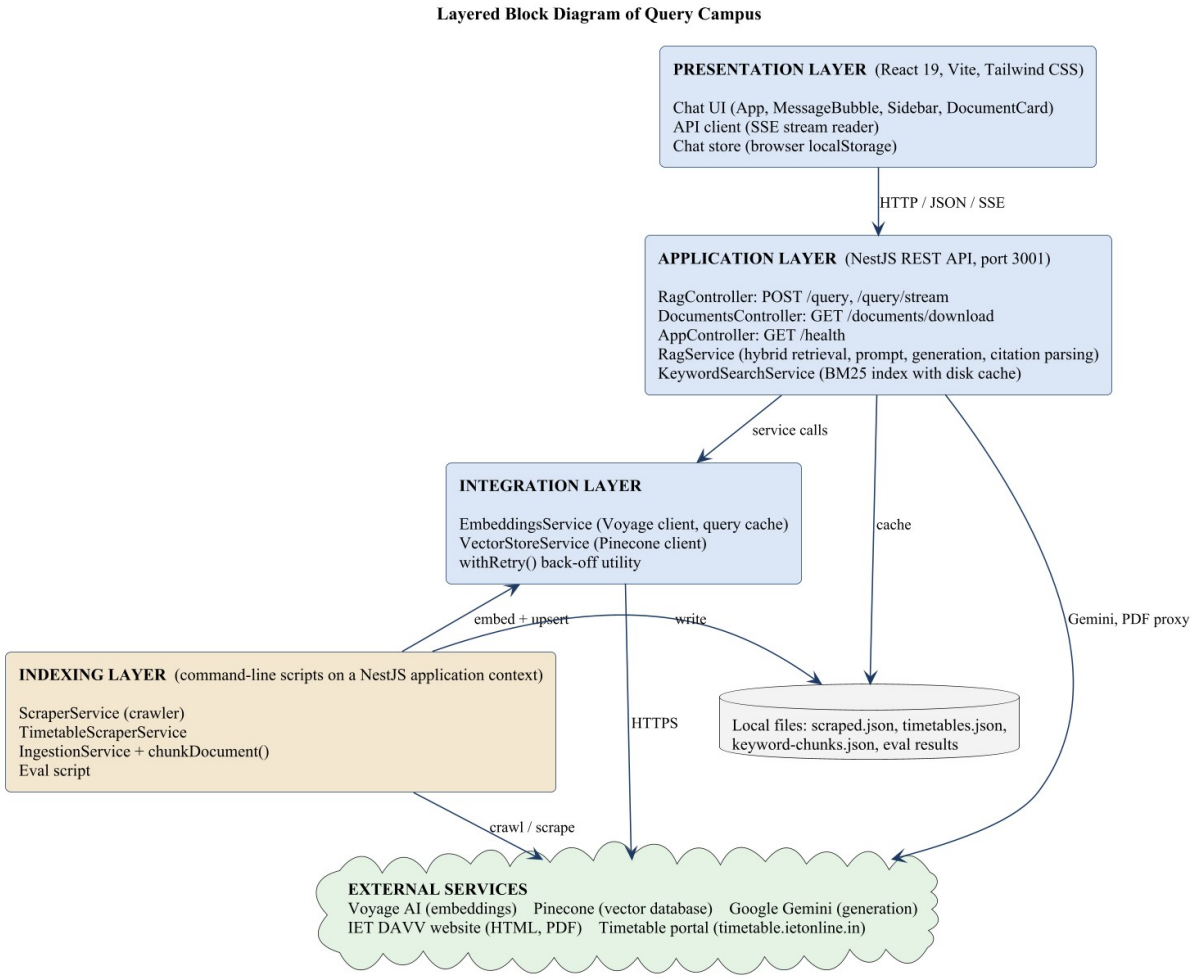


Fig. 3.2: Architecture (block) diagram of Query Campus

Screen layer. The chat screen shows the conversation, sends questions to the backend and saves chats in the browser.

Application layer. The backend receives questions, searches, asks Gemini and sends the answer. It also gives safe PDF downloads and a health check.

Integration layer. Small helpers that talk to Voyage AI and Pinecone and retry when a service is busy.

Indexing layer. Commands that collect the college text, cut it into chunks and save it in Pinecone.

3.4.2 Sequence Diagrams

Fig. 3.3 shows how a question is answered. The backend checks the question, runs the two searches, merges the results and asks Gemini. The words are sent to the screen as they arrive, and at the end the sources and documents are sent.

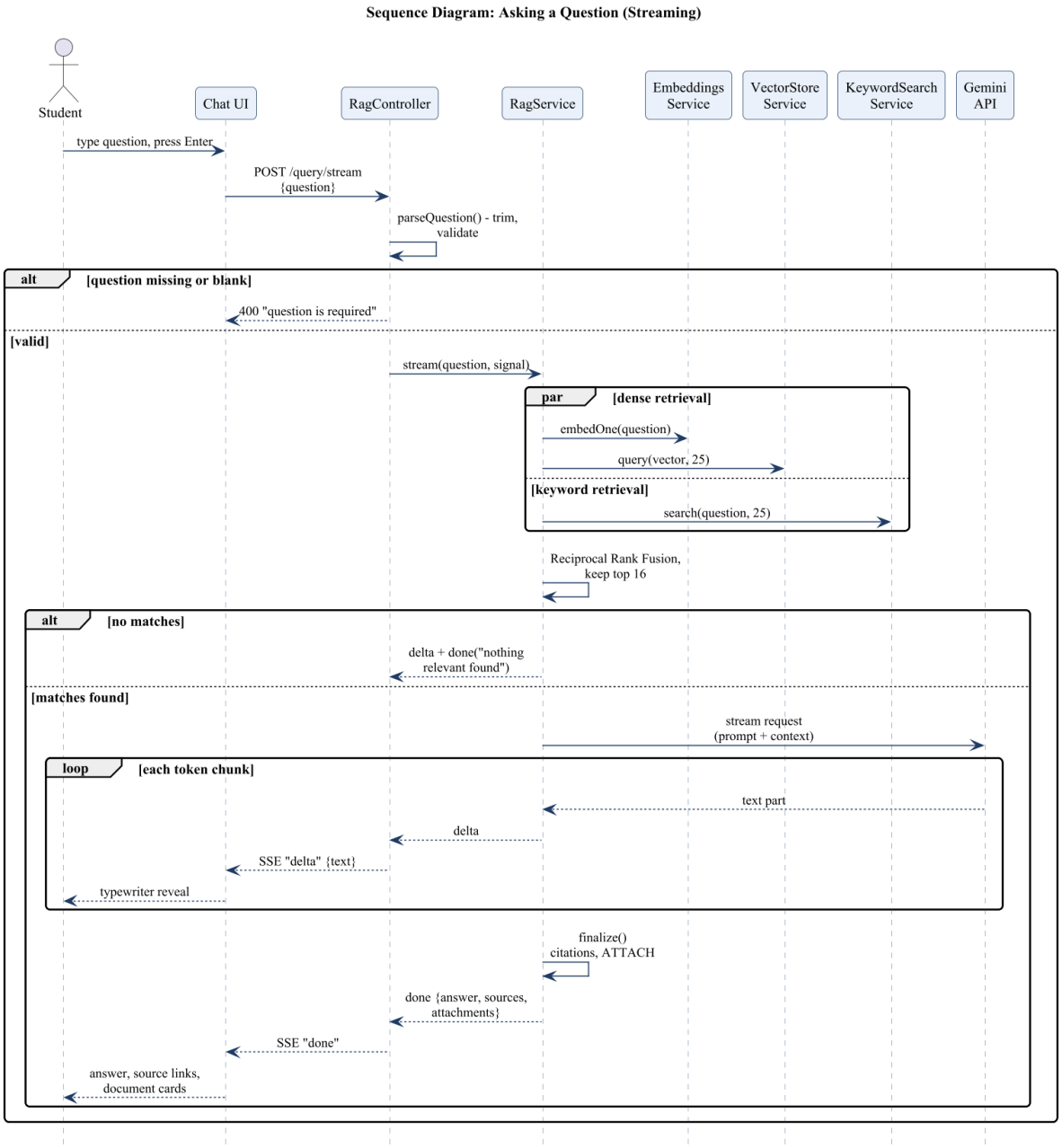


Fig. 3.3: Sequence diagram for asking a question

Fig. 3.4 shows indexing. The operator first collects the pages into a file. Then the system cuts them into chunks, converts a few chunks at a time into numbers, and saves them in Pinecone.

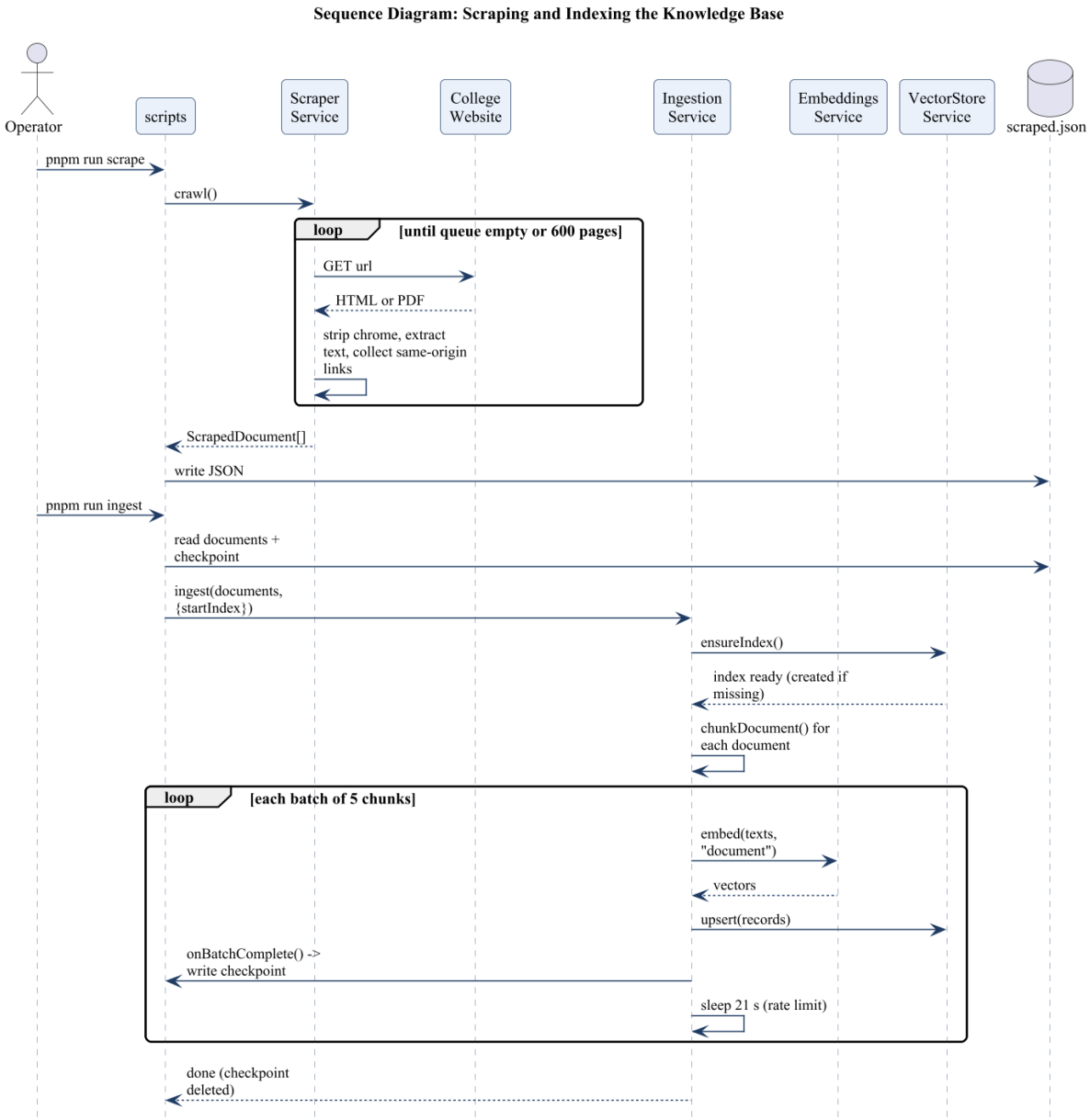


Fig. 3.4: Sequence diagram for scraping and indexing

Chapter-4 Implementation and Testing

4.1 Database Design

Query Campus does not use a SQL database. Its main data is a **vector index in Pinecone**, with a few JSON files on the backend computer. Chat history is kept in the **browser (localStorage)**. Fig. 4.1 shows these entities and how they are related. One document is cut into many chunks. A chat has many messages. A message can have many sources, and each source points to one document. The tables after the figure give the fields of each entity.

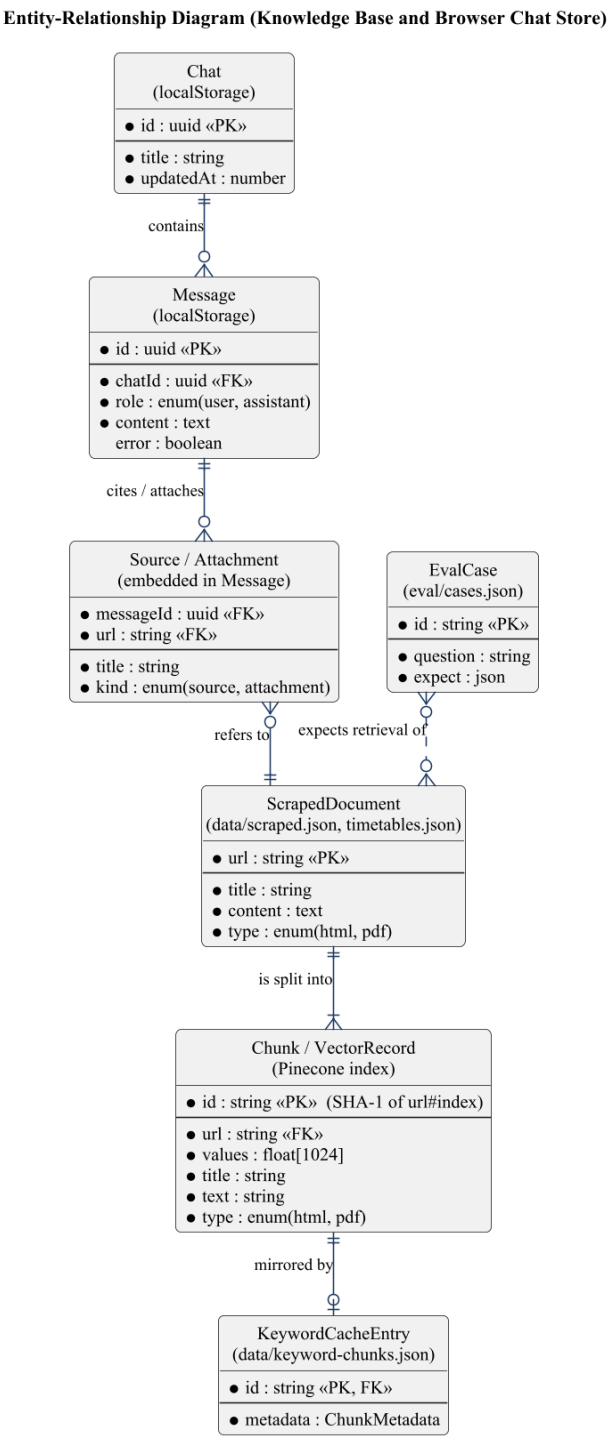


Fig. 4.1: Entity-relationship diagram of Query Campus

Table 4.1: ScrapedDocument (scraped.json and timetables.json)

Field	Type	Key	Description
url	string	PK	Address of the page or file.
title	string		Page title or file name.

Field	Type	Key	Description
content	text		The cleaned text.
type	enum		“html” or “pdf”.

One entry for every collected web page, PDF or timetable document.

Table 4.2: Chunk (Pinecone index “query-campus”)

Field	Type	Key	Description
id	string	PK	A code made from the page address and the chunk number, so re-indexing does not create duplicates.
values	float[1024]		The numbers that represent the meaning of the text.
url	string	FK	Address of the source document.
title	string		Title of the source document.
text	string		The chunk text.
type	enum		“html” or “pdf”.

One record for each chunk of up to 1,500 characters. The index has 2,864 records, of which 350 are timetable chunks (ids start with “tt:”).

Table 4.3: KeywordCacheEntry (keyword-chunks.json)

Field	Type	Key	Description
id	string	PK, FK	Same as the chunk id.
metadata	object		Address, title, text and type of the chunk.

A copy of every chunk on disk, so the keyword search can start quickly.

Table 4.4: Chat (browser localStorage)

Field	Type	Key	Description
id	uuid	PK	Unique code of the chat.
title	string		The first question, shortened to 60 characters.
updatedAt	number		Time of the last change.
messages	Message[]		The messages of the chat.

One conversation.

Table 4.5: Message (stored inside a Chat)

Field	Type	Key	Description
id	uuid	PK	Unique code of the message.
role	enum		“user” or “assistant”.
content	text		The question or the answer.
sources	RagSource[]	FK	Pages used for the answer (address, title, type).
attachments	RagSource[]	FK	Documents the user asked for.
error	boolean		True for an error message.

One question or one answer. Sources and attachments point to documents by address.

Table 4.6: EvalCase (eval/cases.json)

Field	Type	Key	Description
id	string	PK	Short name of the test.
question	string		The question to ask.
expect	object		The page to find, the words the answer must contain, and whether a document must be attached.

One test question with what a correct answer must contain.

4.2 Class Diagram

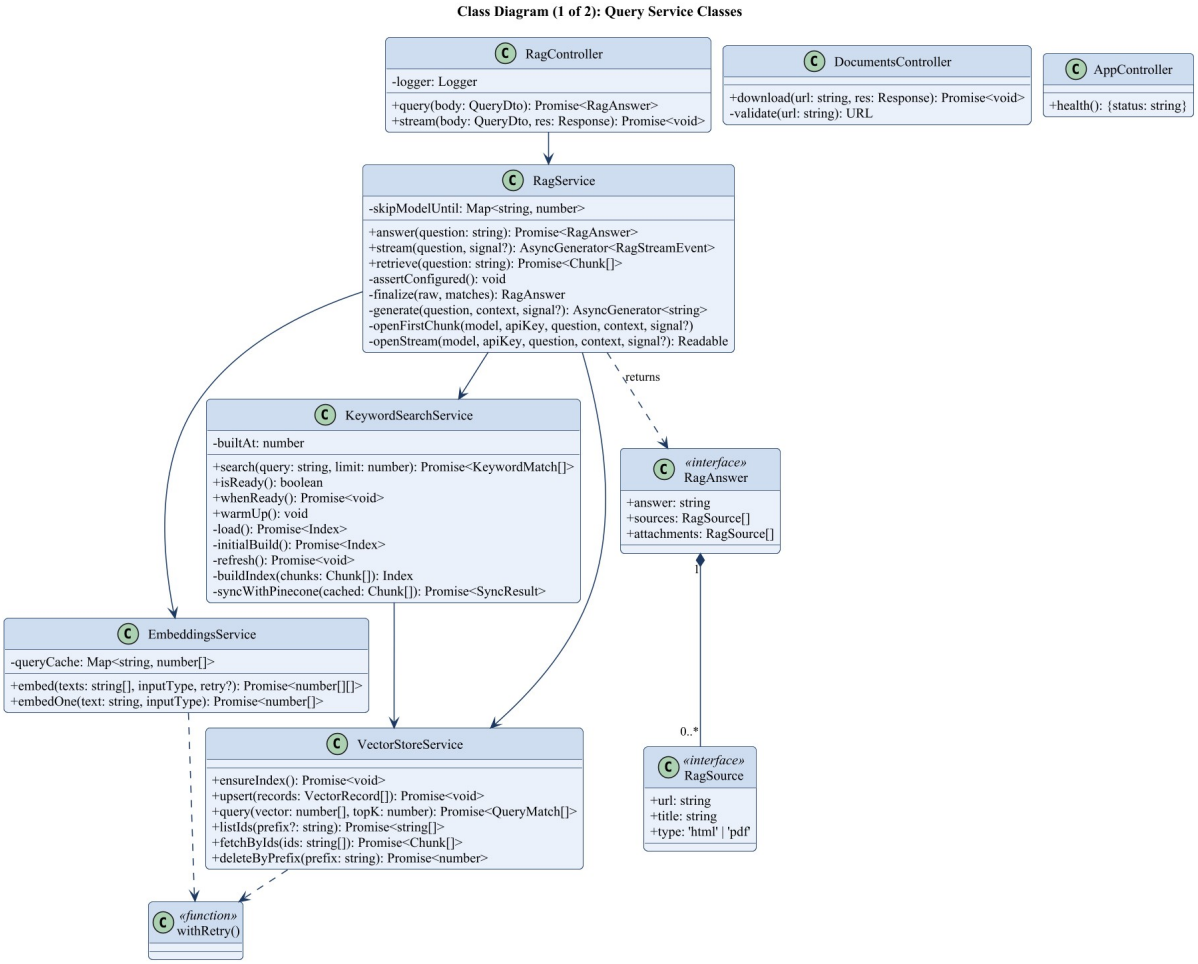


Fig. 4.2: Class diagram (1 of 2): query service classes

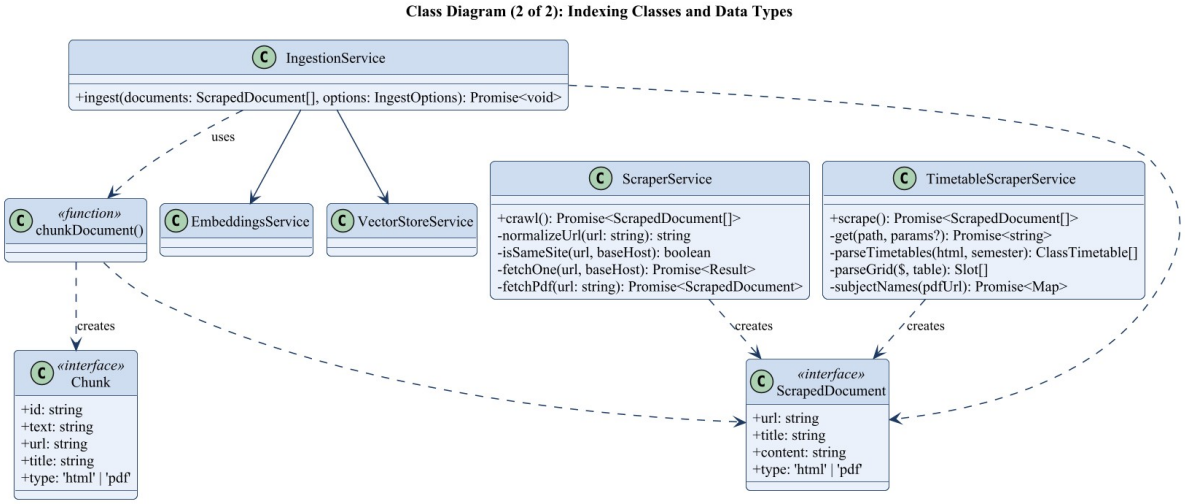


Fig. 4.3: Class diagram (2 of 2): indexing classes and data types

Explanation of the classes

This part explains what each class in Figs. 4.2 and 4.3 is for and what its main methods do, with the parameters they take.

4.2.1 RagController

Purpose. Receives questions from the screen and sends back the answer.

Method	Parameters	What it does
query	body: { question }	Checks the question (it must be text, not blank, and at most 500 characters) and returns the whole answer as JSON.
stream	body: { question }, res	Checks the question, then sends the answer to the screen in small parts as it is written. If the user closes the page, it stops.

4.2.2 DocumentsController

Purpose. Lets users download college PDF files.

Method	Parameters	What it does
download	url, res	Gets the PDF from the college website and sends it to the user as a download.
validate	url	Accepts only PDF files from the college website, so the route cannot be misused.

4.2.3 AppController

Purpose. Tells whether the server is running.

Method	Parameters	What it does
health	none	Returns {status: “ok”}.

4.2.4 RagService

Purpose. The main class. It finds the best text, asks Gemini for an answer and prepares the final answer.

Method	Parameters	What it does
answer	question	Returns the complete answer in one piece.
stream	question, signal	Finds the chunks, asks Gemini, and gives the words one by one, then the final answer with sources.
retrieve	question	Searches by meaning and by keywords at the same time and merges the results. It puts faculty pages first for questions about a person, keeps at most 3 chunks per page, and returns the best 16. If one search fails, the other is used.
generate	question, context, signal	Sends the question and chunks to Gemini. If a model is busy or slow, it tries the next model. It gives up after 40 seconds.
finalize	raw text, chunks	Turns citation numbers like [2] into source links and finds which documents the user asked for.

4.2.5 KeywordSearchService

Purpose. Searches for exact words such as names and codes.

Method	Parameters	What it does
search	query, limit	Finds chunks that contain the words of the question. It also understands “2nd year IT” as “II Year Information Technology”.
isReady	none	Tells whether the keyword index is ready.
whenReady	none	Waits until the keyword index is ready.
warmUp	none	Builds the index in the background when the server starts.

Method	Parameters	What it does
refresh	none	Compares the saved copy with Pinecone and adds new chunks.

4.2.6 EmbeddingsService

Purpose. Turns text into numbers using Voyage AI.

Method	Parameters	What it does
embed	texts, inputType, retry	Sends texts to Voyage AI and returns their numbers. It retries when the service is busy.
embedOne	text, inputType	Does the same for one text. Questions are remembered and are tried only once, so the system can switch to keyword search at once.

4.2.7 VectorStoreService

Purpose. Talks to the Pinecone database.

Method	Parameters	What it does
ensureIndex	none	Creates the index if it does not exist.
upsert	records	Saves records in groups of 100.
query	vector, topK	Returns the topK closest records.
listIds	prefix	Lists the ids of all records.
fetchByIds	ids	Gets records by id.
deleteByPrefix	prefix	Deletes all records whose id starts with the prefix (used for timetables).

4.2.8 IngestionService

Purpose. Saves collected documents in the database.

Method	Parameters	What it does
ingest	documents, options	Cuts documents into chunks, optionally skips chunks that are already saved, converts a few chunks at a time into numbers and saves them. Batch size and waiting time can be set.

4.2.9 ScraperService

Purpose. Collects text from the college website.

Method	Parameters	What it does
crawl	none	Visits pages one after another, stays on the college website, and returns the text of every page and PDF.
fetchOne	url, baseHost	Reads one page: removes menus and footers, keeps the main text and collects the links.
fetchPdf	url	Downloads one PDF and reads its text.

4.2.10 TimetableScraperService

Purpose. Collects the class timetables from the timetable website.

Method	Parameters	What it does
scrape	none	Reads all timetables of the latest session and makes one document per class and per teacher.
parseTimetables	html, semester	Reads the classes, rooms, coordinators and weekly slots from a page. Empty entries are skipped.

4.2.11 Helper functions

Purpose. Small functions used by several classes.

Method	Parameters	What it does
chunkDocument	doc, idPrefix	Cuts a document into chunks of 1,500 characters with 200 characters of overlap.
withRetry	fn, options	Runs a function again after a wait when an online service is busy.
normalizeText	text	Makes “time table” and “timetable” the same word.
expandQuery	query	Adds the college wording to a question (for example “II Year” for “2nd year”).
limitPerKey	items, key, max	Keeps at most max items for each key.
rateLimit	maxPerMinute	Stops a visitor who sends too many questions in a minute.

Data types. ScrapedDocument (url, title, content, type) is made by the scrapers. Chunk (id, text, url, title, type) is made by chunkDocument. RagSource (url, title, type) describes a page or document. RagAnswer (answer, sources, attachments) is the final result of a question.

4.3 Test Cases

4.3.1 Testing techniques used

- **Black-box testing:** we give an input and check the output, without looking at the code.
- **Invalid input testing:** blank, very long and wrong inputs.
- **Automatic tests:** 13 unit tests and 10 end-to-end tests that run with one command.
- **Accuracy test:** 18 real questions. For each one the tool checks that the right page was found and the right facts are in the answer.
- **Manual testing in the browser** on a computer-size and a phone-size screen, and with the backend switched off.

The tests were run on 22 and 23 October 2026 on the running application, with the final database of 2,864 chunks.

4.3.2 Screen snapshots and test cases

Snapshot 1: Home screen. The home screen has a cream page and a side panel for saved chats. In the middle are the heading “Ask about” with a word that changes every two seconds, four suggestion cards and the question box.

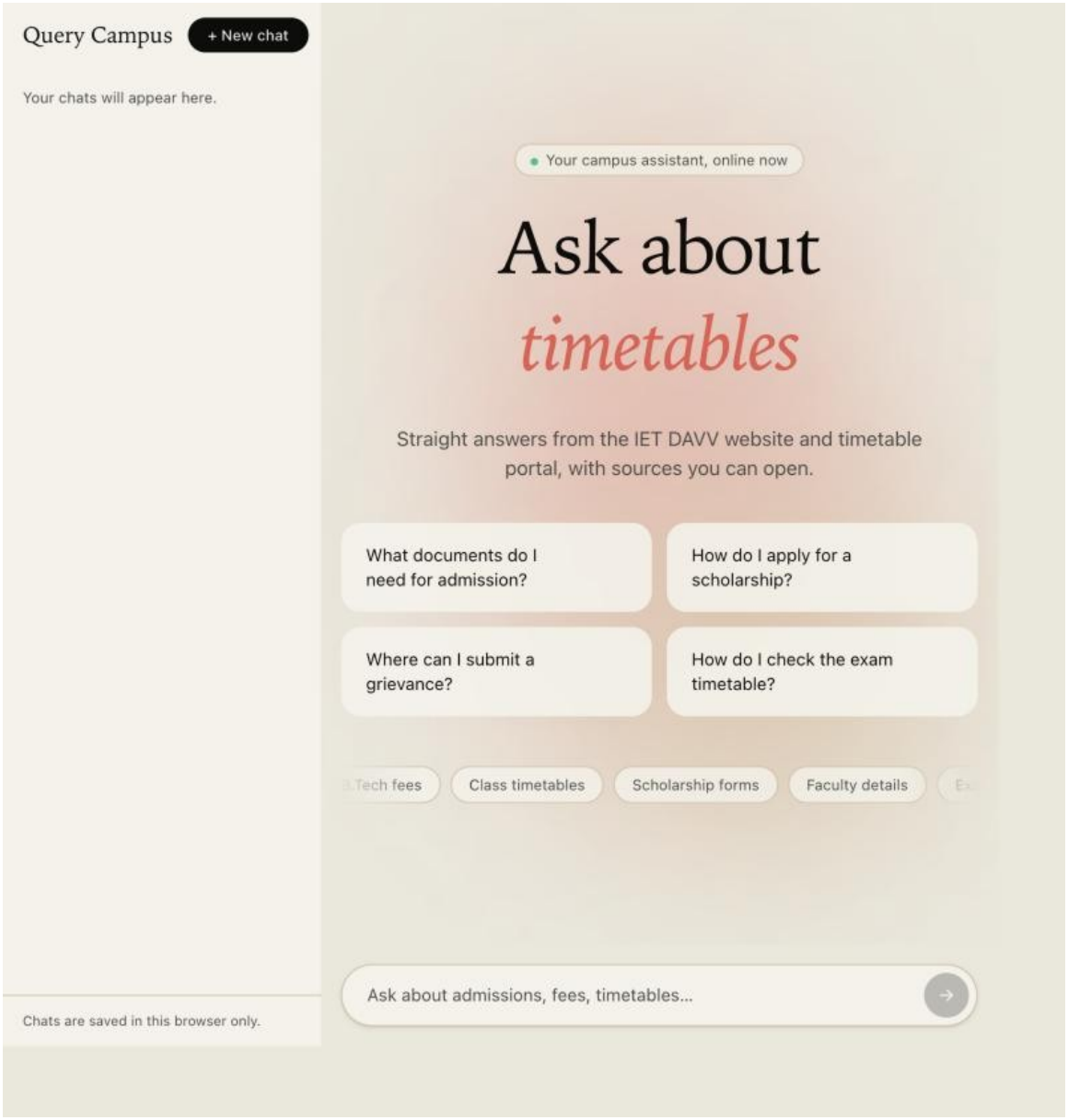


Fig. 4.4: Home screen of Query Campus

Test 01: Test case for opening the website

Input: Open http://localhost:5173 in a browser

System output: Home screen with four suggestion cards and the question box

Result: Passed

Test 02: Test case for the health check

Input: GET /health

System output: HTTP 200, {“status”:“ok”}

Result: Passed

Snapshot 2: A question with an answer. The answer is short and correct, and the page it came from is shown under “Sources”.

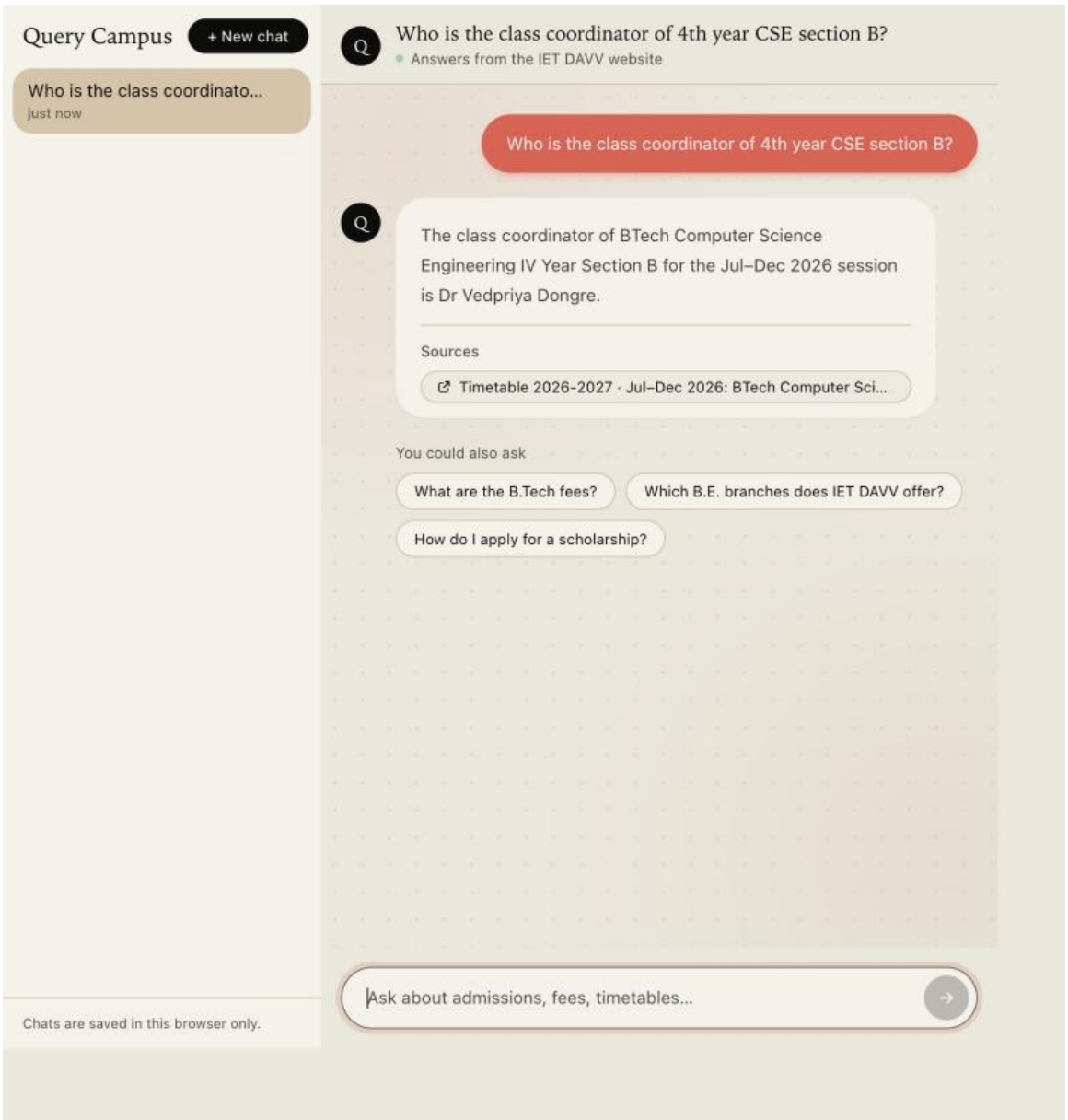


Fig. 4.5: Answer with a source link

Test 03: Test case for a valid question

Input: Who is the class coordinator of 4th year CSE section B?

System output: The class coordinator of BTech Computer Science Engineering IV Year Section B for Jul–Dec 2026 is Dr Vedpriya Dongre. Source: the timetable of that class.

Result: Passed

Snapshot 3: Asking for a document. The answer gives the main details and a document card with an Open button.

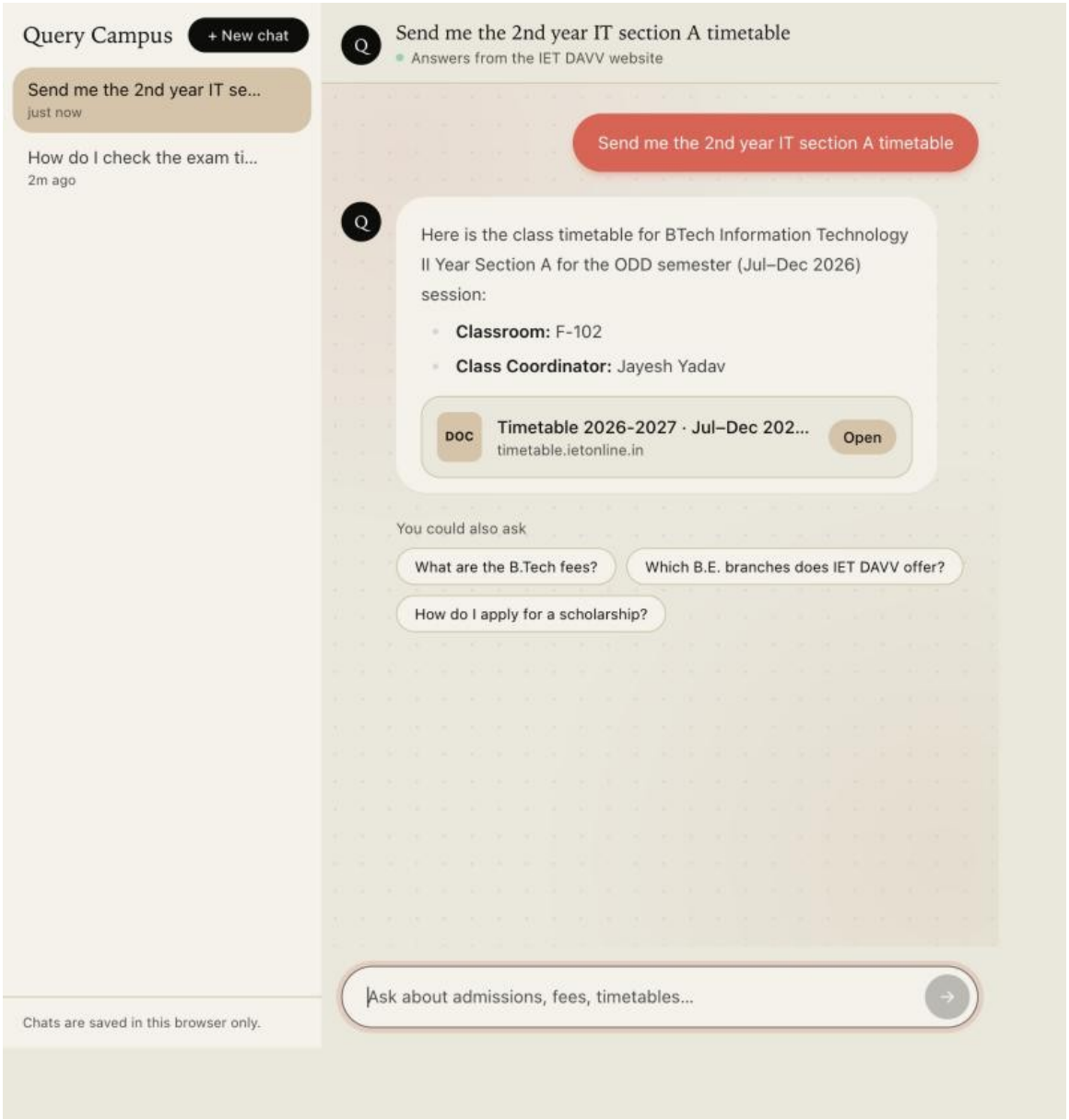


Fig. 4.6: Answer with a document card

Test 04: Test case for a document request

Input: Send me the 2nd year IT section A timetable

System output: Details (BTech Information Technology II Year Section A, classroom F-102, coordinator Jayesh Yadav) and a document card with an Open button

Result: Passed

Snapshot 4: A question that cannot be answered. The assistant says it could not find it. It does not make up an answer.

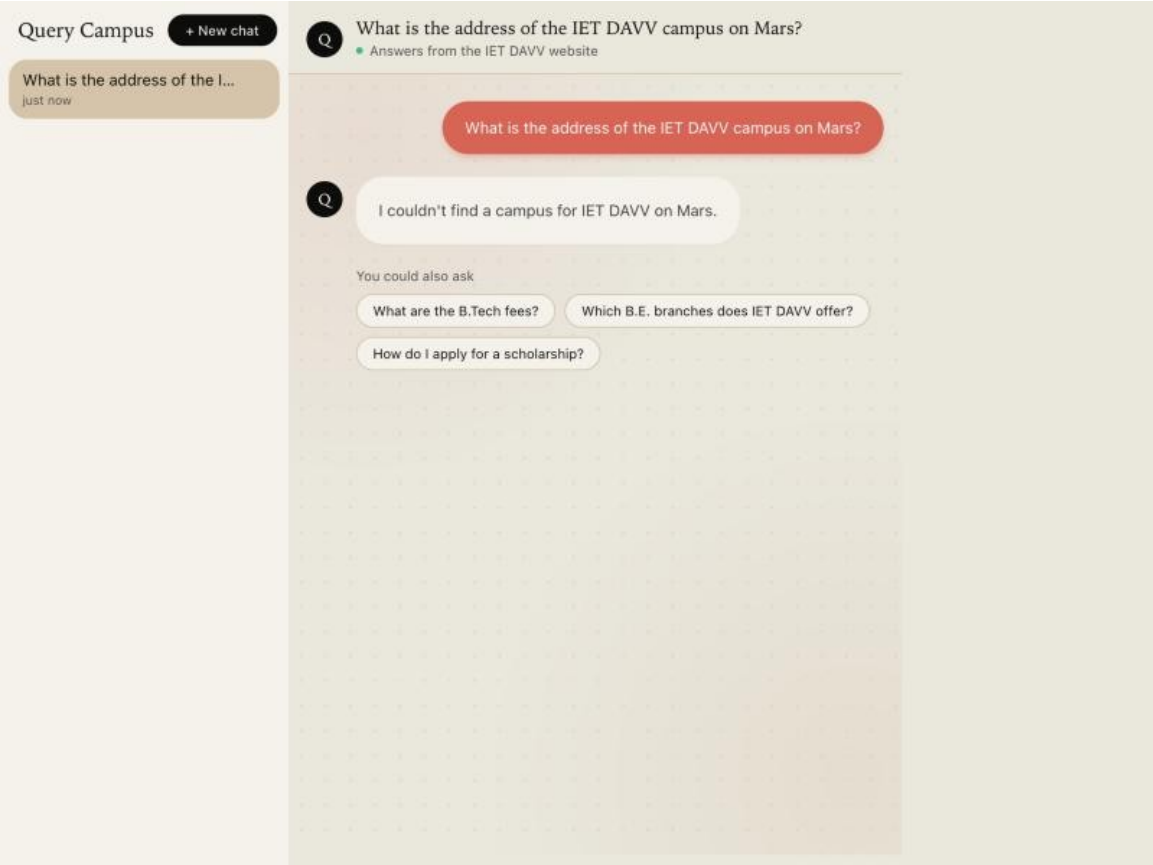


Fig. 4.7: Reply when the information is not available

Test 05: Test case for an unanswerable question

Input: What is the address of the IET DAVV campus on Mars?

System output: I couldn’t find a campus for IET DAVV on Mars. (no sources)

Result: Passed

Test 06: Test case for a question with no exact answer

Input: How do I check the exam timetable?

System output: I couldn't find the exact exam timetable on the college website. You can visit the official website for updates on the exam schedule, class timetable and academic calendar.

Result: Passed

Test 07: Test case for a blank question

Input: POST /query with a question of spaces only

System output: HTTP 400, "question is required"

Result: Passed

Test 08: Test case for a very long question

Input: POST /query with 501 characters

System output: HTTP 400, "question is too long"

Result: Passed

Test 09: Test case for a download from another website

Input: GET /documents/download?url=https://example.com/a.pdf

System output: HTTP 400, "Only PDF documents from the college website can be downloaded"

Result: Passed

Test 10: Test case for too many questions (limit set to 5)

Input: Six questions in one minute from one address

System output: The first five are answered; the sixth gets HTTP 429 and "Too many questions. Please wait a minute and try again."

Result: Passed

Test 11: Test case for a stopped backend

Input: Ask a question while the backend is off

System output: An error message: "Couldn't reach the server. Please check your connection and try again in a moment."

Result: Passed

4.3.3 Accuracy test results

Table 4.7 lists the 18 test questions. **All 18 questions passed.** The right page was found in all 14 questions that name a page, and it was first in 8 of them. A normal answer takes 1.5 to 4 seconds. When Gemini is busy, a backup model answers and it can take up to about 36 seconds.

Table 4.7: Accuracy test of 18 questions (23 October 2026)

No.	Question	Result	Rank of right page	Answer time
1	Who is Rohit Pathak?	Pass	4	3.8 s
2	What does Rohit Pathak teach?	Pass	1	1.5 s
3	Which classes does Vedpriya Dongre teach?	Pass	1	6.9 s
4	Who is the class coordinator of 4th year CSE section B?	Pass	1	3.3 s
5	What is the research interest of Dr. Sufia Aziz?	Pass	2	1.9 s
6	Who is the HOD of Applied Science?	Pass	1	1.9 s
7	Tell me about Hemant Makwana	Pass	2	2.7 s
8	Who is Ashesh Tiwari?	Pass	8	4.7 s
9	Who is Vijay Kumar Karma?	Pass	2	3.9 s
10	Who is Ravi Sindal?	Pass	2	2.3 s
11	What is Lalit Gehlod's designation?	Pass	1	36.2 s (model busy)
12	What are the B.Tech fees?	Pass	1	5.2 s
13	Which B.E. branches does IET DAVV offer?	Pass	n/a	2.8 s
14	Send me the 4th year CSE section B timetable	Pass	1	4.0 s
15	Send me the 2nd year IT timetable	Pass	n/a	3.1 s

No.	Question	Result	Rank of right page	Answer time
16	What is the address of the IET DAVV campus on Mars?	Pass	n/a	1.5 s
17	How do I check the exam timetable?	Pass	n/a	20.2 s (model busy)
18	Where can I see the exam results?	Pass	1	4.9 s

4.3.4 Improvements made after testing

- **More pages.** The first crawl stopped at 600 pages. After raising the limit, 202 more documents were added (2,864 chunks in total).
- **Better keyword search.** Different spellings like “time table” and “timetable” now match, and “2nd year IT” also searches “II Year Information Technology”.
- **Better results.** One page can fill at most three of the 16 result slots, and faculty pages come first for “Who is…?” questions.
- **More reliable.** If one search is busy, the other one answers. The server also stops waiting after 40 seconds.
- **Safe for the public.** A limit on questions per visitor, a limit on question length and a list of allowed websites were added.

Chapter-5 Conclusion

What we learned. We learned how a complete RAG system is built and where it can fail. We learned to get clean text from web pages and PDFs, to cut text into chunks, and to use embeddings and a vector database. The biggest lesson was that good search matters more than the AI model. Search by meaning alone missed names and numbers, and adding keyword search fixed that. We also learned to handle slow or limited online services with retries, backup models and time limits, to stream answers, and to test with a fixed set of questions. We also practised NestJS, React, Tailwind CSS and UML diagrams.

Key results. Table 5.1 sums up the results of the project.

Table 5.1: Key results

Item	Result
Documents in the database	1,009 (160 web pages, 606 PDFs, 243 timetable documents)
Chunks in the database	2,864
Accuracy test	All 18 questions passed
Normal answer time	1.5 to 4 seconds
Automatic tests	13 unit tests and 10 end-to-end tests, all passed
Visitor limit	20 questions per minute for each visitor

What we achieved.

- A database of 1,009 documents (160 web pages, 606 PDFs and 243 timetable documents) in 2,864 chunks.
- Answers with sources in about one to four seconds, and a clear “I couldn’t find it” when the answer is not available.
- A simple, light chat screen with saved chats, source links, document cards and a phone-friendly layout.
- A backup when Gemini is busy, and keyword-only search when the embedding service is busy.
- Easy updating with two commands, and an accuracy test where all 18 questions pass.

- A website that is ready to deploy, with a limit on questions per visitor and a list of allowed websites.

How to keep the system up to date. The college website changes, so the database must be updated. At the start of every semester, run the timetable command. When the college adds new notices, run the scrape command and then the ingest command with --incremental, which adds only the new pages. After any change to the search or the prompt, run the accuracy test to check that all questions still pass.

Limitations.

- **No chat memory.** Each question is answered on its own, so a follow-up like “and his email?” is not understood.
- **Data can be old.** The data is as new as the last crawl. The system cannot know if the college website is out of date (the college’s own exam timetable page is empty at the moment).
- **Scanned PDFs are not read.** PDFs that are only pictures have no text, and we do not use OCR.
- **Free plan limits.** The free Voyage AI plan allows only 3 requests a minute for all visitors together. When it is reached, the system uses keyword search only, which is a little less exact. A paid plan removes this limit.
- **English only, no login.** Everyone gets the same answers, and users cannot rate answers.

Future work.

- Add chat memory so that follow-up questions work.
- Use a paid embedding plan for faster indexing and no limit.
- Crawl automatically and update only the pages that changed.
- Add OCR for scanned notices, and support for Hindi.
- Add thumbs up and down buttons to collect feedback.
- Add a WhatsApp or Telegram bot, and reuse the system for other colleges.

Conclusion. Query Campus shows that a small, low-cost and self-hosted system can turn a large and hard-to-use college website into a simple chat service. Every answer is based on the

college’s own pages, shows its sources, and says “I couldn’t find it” when the answer is not there. This makes it more reliable than manual browsing or a general chatbot.

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APPENDIX

A. Abbreviation List

Abbreviation	Meaning
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
CSE / IT	Computer Science Engineering / Information Technology
DAVV	Devi Ahilya Vishwavidyalaya
IET	Institute of Engineering & Technology
JSON	JavaScript Object Notation
LLM	Large Language Model
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
SRS	Software Requirements Specification
UI / UML	User Interface / Unified Modeling Language
URL	Uniform Resource Locator

B. Glossary

Term	Meaning
Chunk	A small piece of a document (about 1,500 characters).
Embedding	A list of numbers that stands for the meaning of a text.
Grounding	Making the AI answer only from the given text, with sources.
Hallucination	A confident answer from an AI that is wrong or made up.
Indexing	Cutting documents into chunks and saving them in the database.
Session	A college term such as Jul–Dec 2026.

C. User Manual

C.1 For students, applicants and staff

1. Open the Query Campus website in any browser.
2. Type a question in simple English, or click one of the suggestion cards, and press Enter.
3. Read the answer. Click a link under “Sources” to open the original page.
4. To get a document, ask for it, for example “Send me the 3rd semester syllabus”. Use Open or Download on the document card.
5. Use “+ New chat” to start a new chat, click an old chat to open it, or click the cross to delete it. Chats are saved only in your browser.

C.2 For the operator

1. Run “pnpm install” in the project folder.
2. Copy .env.example to .env in both apps and fill in the API keys (Gemini, Voyage AI, Pinecone).
3. In apps/backend run “pnpm run scrape”, then “pnpm run ingest”. Use “pnpm run ingest --incremental” to add only new pages.
4. Run “pnpm run timetables” at the start of every semester.
5. Start the app with “pnpm run dev:backend” and “pnpm run dev:frontend”.
6. After changing the prompt or the search, run “pnpm run eval” and check that nothing got worse.

D. Main Settings

Setting	Default	Meaning
PORT	3001	Backend port
CORS_ORIGIN	any	Website allowed to call the API (set it in production)
RATE_LIMIT_PER_MINUTE	20	Questions per visitor per minute

Setting	Default	Meaning
GEMINI_API_KEY	(secret)	Key for Google Gemini
VOYAGE_API_KEY	(secret)	Key for Voyage AI
PINECONE_API_KEY	(secret)	Key for Pinecone
VITE_API_URL	http://localhost:3001	Backend address used by the website

E. Deployment

1. Backend on a Node host (for example Render or Railway): run “pnpm install”, “pnpm --filter backend build” and “pnpm --filter backend start:prod”.
2. Set the three API keys, CORS_ORIGIN (your website address) and TRUST_PROXY=1 if the host uses a proxy.
3. Fill the Pinecone database once: run the scrape and ingest commands from your computer.
4. Frontend on a static host (for example Netlify or Vercel): build with VITE_API_URL set to the backend address and upload apps/frontend/dist.
5. Open /health on the backend and ask a few questions on the website to check that everything works.